MILITARY SPECIFICATION

TIMER, SEQUENTIAL: M16 AND M30

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of mechanical timer entitled Timer, Sequential, Training, M16 and M30 which is used with the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8800667 - Timer Sequential Training: M16.

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8821960 - Base Assembly Training
8830832 - Timer Sequential Training: XM30.
8862385 - Base and Movement Mounting Plate, Assembly, Training.
EL-8885951 - Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The timer shall comply with all requirements specified on the drawing (dwg) 8800667 and 8830832, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Electrical continuity. The timer shall show electrical continuity between the pins of the connector when tested as specified 4.4.1.

3.5 Insulation resistance. The insulation resistance between electrically isolated circuits and between ungrounded circuits shall be not less than 20 megohms when subjected to a direct current (dc) potential of 500 ± 50 volts when tested as specified in 4.4.2.

3.6 Operation.

3.6.1 Setting torque. The torque required to set the timer shall not exceed 15 inch pounds and the readout dial of the assembly shall creep less than one digit after the assembly has been set and the wrench removed when tested as specified in 4.4.3.1.

3.6.2 Setting. The timer shall be capable of being set at any position between 0.000 and 129.999 seconds when tested as specified in 4.4.3.2.

3.7 Pressure retention. The timer shall not leak more than 0.5 pounds per square inch pressure in 15 minutes when subjected to a pressure of 10 to 12 pounds per square inch gage when tested as specified in 4.4.4.

3.8 Environmental testing. The timer shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The timer shall also comply with 3.4, 3.5 and 3.6 before and after each environmental test. Testing shall be as specified in 4.4.5. Tests shall be conducted in the order specified.

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3.8.1 Vibration. The timer shall be capable of withstanding vibration in any direction without evidence of damage such as loose parts and shall comply with the requirements of 3.4, 3.5 and 3.6 after being tested as specified in 4.4.5.1.

3.8.2 Shock. The timer shall withstand without damage, a shock in any direction with a peak amplitude of 20 + 2 g's and shall comply with the requirements of 3.4, 3.5 and 3.6 after being tested as specified in 4.4.5.2.

3.9 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.10 Workmanship.

3.10.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.10.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.10.3 Burrs. Parts shall be free from burrs.

3.10.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.10.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8800667 and 8830832) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

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Table I
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Electrical continuity (see 3.4)		X
3. Insulation resistance (see 3.5)		X
4. Operation (see 3.6)		X
5. Vibration (see 3.8.1)	X	
6. Shock (see 3.8.2)	X	
7. Pressure retention(see 3.7)		X

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristic in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Base assembly training (see dwg 8821960 covering a detail of dwg 8800667) Base and movement mounting plate, assembly, training (see dwg 8862385 covering a detail of 8830832).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Wire broken or damaged	Visual
	102. Cold solder connection	Visual
	103. Connector pin damaged	Visual
	104. Gear damaged	Visual
	105. Gear teeth improperly aligned	Visual
	106. Hardware loose or missing	Visual
	107. Improper soldering (see 3.9)	Visual
	108. Objective evidence relating to material missing or improper (see 3.1)	Visual

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Minor:
201. Evidence of poor workmanship (see 3.10).....Visual

4.3.2.1.2 Fuze, sequential timer, training: M16 and M30 (see dwg 8800667 and 8830832).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Cover improperly assembled	Visual
	102. Connector pin damaged	Visual
	103. Marking incorrect, mislocated or unidentifiable	Visual
	104. Hardware loose or missing	Visual/Manual
	105. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.3 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation, packing and packaging not in accordance with MIL-STD-1169	Visual
	102. Marking not in accordance with MIL-STD-1169	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test classification
a.	Electrical continuity	Major
b.	Insulation resistance	Major
c.	Operation	Major

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d. Vibration — Major
e. Shock — Major

Of these tests, d and e are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Electrical continuity, insulation resistance and operation. Each timer in the lot shall be tested in accordance with 4.4.1, 4.4.2 and 4.4.3. If any item fails to comply with 3.4, 3.5 and 3.6, the item shall be rejected and removed from the lot.

4.3.3.2 Vibration and shock. One sample timer shall be selected at random for each twenty five timers or portion thereof produced. The sample shall be tested in accordance with 4.4.5.1 and 4.4.5.2. If any item fails to comply with 3.8.1 and 3.8.2, the lot shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8885951). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Electrical continuity. The electrical resistance measurement between the following pin pairs of the electrical receptacle shall not exceed 0.030 ohm.

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From Pin	To Pin
A	E
B	D, F
C	Y
H	K, M
J	L, N
P	S
R	T, V
U	W
X	Z
AA	CC
BB	DD

4.4.2 Insulation resistance. Each connector contact listed in Table IV shall be subjected to a dc potential until a steady reading is obtained, using an insulation resistance tester from which a direct reading in megohms can be made.

Table IV

Pin energied	All other pins to ground except
A	E
B	D, F
C	Y
H	K, M
J	L, N
P	S
R	T, V
U	W
X	Z
AA	CC
BB	DD

4.4.3 Operation.

4.4.3.1 Setting torque. The breakaway and maintaining torque necessary to set the unit shall be measured with a torque wrench. The dial shall not move a measurable distance after the setting torque is removed.

4.4.3.2 Setting. The timer shall be set at various dial settings when a torque not exceeding 15 inch pounds is applied by a torque wrench.

4.4.4 Pressure retention. The timer shall be subjected to a pressure differential not to exceed 12 pounds per square inch (p.s.i.) nor fall below 10 p.s.i. The greater pressure shall be exerted inside the timer.

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The change in pressure inside the timer that occurs while the timer is subjected to the required pressure differential for 15 minutes shall be determined. The 15 minute test interval shall start immediately after the required pressure differential is obtained. Any change in pressure inside the timer during the test interval is a leak which must be measured to determine conformance to the requirements.

4.4.5 Environmental testing (see 3.8).

4.4.5.1 Vibration. The timer shall be mounted to a rigid fixture at station 90.05. The fixture shall be mounted to a suitable vibration producing apparatus. The timer shall then be vibrated through a range of frequencies from 5 to 55 Hertz (Hz) at a constant double amplitude of 0.06 + 0.006 inch for a duration of 30 minutes in each of its three (3) major mutually perpendicular axes. Each cycle 5 to 55 to 5 Hz shall have a duration of one (1) minute and the frequency shall be varied uniformly throughout the cycle. The timer shall show no damage after the vibration test and shall comply with the requirements of 3.4, 3.5 and 3.6.

4.4.5.2 Shock. The timer shall be tested in accordance with MIL-E-5272, Procedure II except that the 15 g's shall be changed to 20 ± 2 g's. After completion of the test the timer shall be tested as specified in 4.4.1, 4.4.2 and 4.4.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The timer shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The timer shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The timer shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

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6.1.2 Contract data requirements (see 4.3.4.1). specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-1807 Revision 1 dated 5 November 1964.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A152

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